

AMAN MODANWAL

Assistant Software Engineer

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Portfolio | LinkedIn | GitHub

PROFESSIONAL SUMMARY

Assistant Software Engineer with **4 years** of experience building production full-stack web and desktop applications using React.js, Node.js, Express.js, JavaScript, TypeScript, HTML5, CSS3, MySQL, and Electron. Shipped **5+ production systems**, **20+ REST APIs**, and real-time solutions (WebSocket/Socket.IO) for railway infrastructure and surveillance, including NVR platforms supporting **64+ concurrent camera streams** and secure content deployment tools. Strong in Linux, Git, Networking (TCP/IP, DNS, DHCP), ONVIF, RTSP, Agile/Scrum, performance optimization, and releasing high-quality maintainable software on time.

TECHNICAL SKILLS

Frontend: React.js, JavaScript (ES6+), TypeScript, HTML5, CSS3, SCSS, Redux Toolkit, React Hooks, Context API, Bootstrap, React Bootstrap, Chakra UI, jQuery, Responsive Web Design

Backend: Node.js, Express.js, REST APIs, RESTful API Design, JWT Authentication, WebSocket, Socket.IO, API Integration

Languages: JavaScript, TypeScript, C, Basic Python, Kotlin

Databases: MySQL, MongoDB, Database Design, Query Optimization

Tools & Technologies: Electron, Electron Builder, Vite, Webpack, Babel, Git, GitHub, NPM, Yarn, Postman

Networking & Protocols: TCP/IP, DHCP, DNS, RTSP, ONVIF, SOAP/XML

Operating Systems & Practices: Linux, Windows, Agile, Scrum, SDLC, Performance Optimization, Code Review

PROFESSIONAL EXPERIENCE

Assistant Software Engineer

December 2022 - Present

PPS International Pvt Ltd

Greater Noida, India

- Developed and maintained **5+ production full-stack web and desktop applications** (React.js, Node.js, Express.js, Electron, MySQL) for railway infrastructure and surveillance, deployed across multiple train fleets and serving continuous operations.
- Designed and shipped **20+ RESTful APIs** with JWT authentication; optimized queries and caching to cut average API response time by **~30%** and support real-time data processing under load.
- Created a library of **40+ reusable React components** (Hooks, HTML5, CSS3), reducing new UI feature development time by **~25%** and improving consistency across the product suite.
- Implemented WebSocket and Socket.IO real-time layers that achieved **sub-second data synchronization** for live monitoring dashboards used daily by operations teams.
- Integrated ONVIF IP cameras and RTSP streaming into NVR applications, enabling stable concurrent live video from **16+ cameras** with discovery, recording, playback, and configuration.
- Architected the SPUT Electron desktop tool for secure transfer of content packages up to **~10GB** to IoT Web Server boards and onboard infotainment systems, supporting single and multi-device deployment.
- Resolved production defects and performed code reviews in Linux environments, contributing to a **~40% reduction in critical bugs** reported during sprint cycles.
- Completed **100% of assigned sprint features on time** across 30+ Agile/Scrum sprints while collaborating with product, QA, and hardware teams on railway systems.
- Authored technical and API documentation that reduced new developer onboarding time by **~50%** and accelerated knowledge transfer for maintenance and enhancements.

PROJECTS

Network Video Recorder (NVR) - Real-Time Video Monitoring System

Tech Stack: React.js, Node.js, Express.js, REST API, WebSocket, RTSP, ONVIF

- Built a production web-based NVR that monitors and manages ONVIF IP cameras with stable concurrent live streams from **64+ cameras** and low-latency playback.
- Designed scalable REST APIs for camera management and JWT auth; optimized ONVIF (SOAP/XML) + RTSP handling to reduce camera discovery-to-stream latency by **~35%**.
- Launched full camera lifecycle features: discovery, live streaming, recording, playback, and configuration dashboards used daily by operations for continuous surveillance.
- Added WebSocket real-time status updates that cut mean time to detect camera/network failures by **~40%** and improved operator response speed.

SPUT Tool & IOB Server - IP Monitoring, Content Deployment & Train Infotainment Platform

Tech Stack: Electron, React.js, Node.js, Express.js, Offline-First Architecture

- Engineered the SPUT Electron desktop tool to continuously monitor IoT Web Server Board modules via IP pinging, providing real-time health indicators and improving fleet visibility across train systems.
- Implemented secure folder validation and reliable transfer of content packages up to **~10GB** to single or multiple target devices, reducing manual deployment errors by **~60%**.
- Developed a full-stack offline-first IOB Server so passengers can access journey info, entertainment (movies, music, videos, kids zone), tourist places, and railway services in low/no-network conditions.
- Shipped secure authentication and service-request flows fully compatible with SPUT packages, enabling **zero-friction deployment** and cutting content rollout time to onboard systems by **~50%**.

EDUCATION

B.Tech, Computer Science and Engineering	2024 - Present
Dr. A.P.J. Abdul Kalam Technical University, Meerut, India	
Full Stack Development	01/2023 - 08/2023
Apna College (Hybrid), India	
Diploma, Computer Science and Engineering (74%)	2019 - 2022
Board of Technical Education, Jaunpur, India	

CERTIFICATIONS

- Problem Solving (Basic) - HackerRank | DSA with Java - Apna College | Master JavaScript - GeeksforGeeks

LANGUAGES

- Hindi | English